

Problem B. Covered Points Count

Time Limit 3000 ms

Mem Limit 262144 kB

You are given n segments on a coordinate line; each endpoint of every segment has integer coordinates. Some segments can degenerate to points. Segments can intersect with each other, be nested in each other or even coincide.

Your task is the following: for every $k \in [1..n]$, calculate the number of points with integer coordinates such that the number of segments that cover these points equals k . A segment with endpoints l_i and r_i covers point x if and only if $l_i \leq x \leq r_i$.

Input

The first line of the input contains one integer n ($1 \leq n \leq 2 \cdot 10^5$) — the number of segments.

The next n lines contain segments. The i -th line contains a pair of integers l_i, r_i ($0 \leq l_i \leq r_i \leq 10^{18}$) — the endpoints of the i -th segment.

Output

Print n space separated integers $cnt_1, cnt_2, \dots, cnt_n$, where cnt_i is equal to the number of points such that the number of segments that cover these points equals to i .

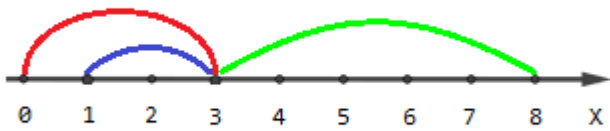
Examples

Input	Output
3 0 3 1 3 3 8	6 2 1

Input	Output
3 1 3 2 4 5 7	5 2 0

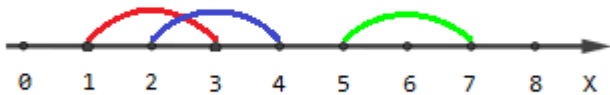
Note

The picture describing the first example:



Points with coordinates $[0, 4, 5, 6, 7, 8]$ are covered by one segment, points $[1, 2]$ are covered by two segments and point $[3]$ is covered by three segments.

The picture describing the second example:



Points $[1, 4, 5, 6, 7]$ are covered by one segment, points $[2, 3]$ are covered by two segments and there are no points covered by three segments.